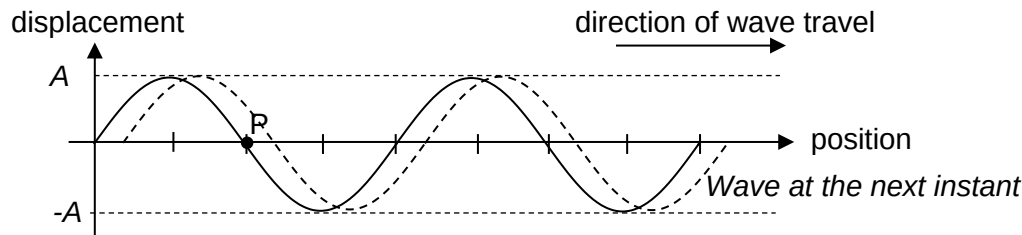


15 C



When the wave profile is drawn in the next instant (dotted line), one can tell that Particle P will be moving upwards in the positive direction, hence option A is incorrect.

As the particle P is undergoing a simple harmonic oscillation, at this instance it is at the equilibrium position, it should have the maximum velocity, zero acceleration. So, options B and D are incorrect.

As the displacement-position graph does not show a decreasing amplitude, all particles along the wave (including Particle P) have an amplitude of A . This is one of the distinguishing features between a progressive wave and a stationary wave. For the latter, the amplitude varies from maximum at the antinode and zero at the node.

16**R**

Resolution or Rayleigh questions necessarily involve small angles